

# ASPERITAS BCR/XAIRA BENCHMARK GPT TRAINING SOURCE

Biodiversity access -> proprietary biological data -> AI-bio models -> DBTL validation -> IP/compliance trust -> products/licensing -> global biological infrastructure

| Field                         | Value                                                                                                                                                                                                                                                                 |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Document type                 | ChatGPT Project Source / Custom GPT Knowledge artifact / AI Lead operating doctrine                                                                                                                                                                                   |
| Version                       | v1.0 KR   Generated 2026-07-07 KST                                                                                                                                                                                                                                    |
| Primary benchmark inputs      | BCR Therapeutics 분석 보고서.pdf; Xaira Therapeutics Analytical Report.pdf                                                                                                                                                                                                 |
| Supporting Asperitas doctrine | ASPERITAS_PROJECT_SOURCE_CONSTITUTION_v11_0_KR.pdf; Asperitas_AI_Lead_Expert_GPT_Training_Source_v1_0_KR.pdf; AI Agent Development Control Plane; IR/public positioning                                                                                               |
| Classification                | Internal / Confidential. Do not paste confidential text, unreleased source data, partner contracts, specimen-level records, or unpublished lab results into external tools without human approval.                                                                    |
| Boundary                      | This PDF is an operating doctrine and benchmark translation. It is not evidence that Asperitas has production vector DB/RAG/KG/eval suite/legal review/wet-lab validation/regulatory approval/autonomous lab execution/foundation model capability already completed. |

**Critical Non-Confusion Rule**

BCR/Basecamp and Xaira are benchmark sources, not Asperitas implementation proof. Use them to shape decisions, source registry, RAG/KG/eval requirements, and claim boundaries. Never convert benchmark ambition into present-tense Asperitas capability without implementation logs, validated data, counsel review, or wet-lab/partner evidence.

## 0. How to Use This PDF as Project Source

Attach this PDF to ChatGPT Project Sources or Custom GPT Knowledge. The model should treat it as an operating doctrine and benchmark memory layer. It should not treat the document as proof of any production system or legal/scientific validation.

| When retrieved                                 | Model behavior                                                                                                                                         |
|------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategy / investor / CEO questions            | Use BCR/Xaira as benchmark archetypes, then translate to Asperitas with Scalability, Moat, Biosafety/Compliance filters.                               |
| RAG/KG/eval / agent build questions            | Prioritize source registry, approved-only ingestion, citation fidelity, schema-first answers, evals, trace logs, and human approval gates.             |
| Biology / DBTL / wet-lab questions             | Stay at strategic and validation-framework level unless approved protocols and biosafety gates exist. Avoid executable high-risk biological protocols. |
| Claims / IR / website / partner deck questions | Separate verified fact, inference, unverified plan, and prohibited overclaim. Attach source_id, confidence, compliance tag, and decision implication.  |

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# 1. Executive Memory Layer

Bottom line: the combined lesson from BCR/Basecamp and Xaira is that AI-bio advantage is not model-only. It compounds when legally traceable proprietary data, biological validation loops, IP/compliance trust, and narrow commercialization modules reinforce each other.

## Five final lessons from BCR + Xaira

- 1. **Data rights beat data volume alone.** BCR is strongest where biodiversity-derived data is traceable and benefit-sharing-aware; Xaira is strongest where perturbation data is generated at scale and linked to models.
- 2. **Foundation-model narrative must be earned.** Both companies frame models as central, but the real defensibility comes from data provenance, assay feedback, partner access, and product translation.
- 3. **Preclinical platform does not equal clinical validation.** Neither benchmark should be read as de-risked clinical success unless named assets, INDs, trial registrations, and clinical results are public and verified.
- 4. **Vertical integration is powerful but expensive.** Xaira shows the high-capital version; Asperitas should start with a narrower vertical wedge before trying to copy the full-stack structure.
- 5. **Trust infrastructure is part of the moat.** Nagoya/CITES/DSI/IP traceability, source registry, claim records, and compliance gates convert biology data into investable infrastructure.

## Five things Asperitas should adopt now

- 1. Build a benchmark source registry first: BCR, Xaira, official pages, papers, patents, datasets, press releases, and legal/compliance artifacts must enter as source candidates with review status.
- 2. Convert biodiversity access into structured data products: specimen/species/provenance/permit/assay metadata, not just narrative claims.
- 3. Select one narrow biological data product or discovery wedge for the first 90 days; avoid claiming a broad foundation model before data/eval/validation evidence exists.
- 4. Create a claim verification record for every IR, website, partner, and CEO-facing claim.
- 5. Build RAG/KG/eval/control plane as operational infrastructure before autonomous agents or multi-agent complexity.

## Five things Asperitas should not copy

- 1. Do not copy Xaira-level burn or vertical integration without capital, assay capacity, executive depth, and a named product wedge.
- 2. Do not call biodiversity access a data moat until rights, provenance, metadata quality, and reuse permissions are structured.
- 3. Do not claim clinical or therapeutic readiness from benchmark analogies, preprints, or model outputs.
- 4. Do not ingest external papers, patents, or web sources into RAG without registry status, license/confidentiality checks, and citation-fidelity evals.
- 5. Do not automate wet-lab protocol design/execution or high-risk biological workflows without biosafety/compliance/human approval gates.

# 2. Source Status / Assumptions

| Source group                                | Status in this PDF                                                                                              | Use                                                                                                               | Caveat                                                                                                 |
|---------------------------------------------|-----------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| BCR Therapeutics 분석 보고서.pdf                 | Active benchmark source; official claims inside remain source candidates until separately reviewed.             | Biodiversity data moat, BaseData/EDEN, staged therapeutic platform, partnership/IP lessons.                       | Not proof of clinical success; several performance claims are press/preprint/company-claim level.      |
| Xaira Therapeutics Analytical Report.pdf    | Active benchmark source; 2026 items treated as candidate facts until official URLs are reviewed at time of use. | Capital-intensive vertical integration, perturbation data flywheel, RFdiffusion/RFantibody, X-Atlas/X-Cell model. | Not proof of de-risked pipeline; named assets/IND/trials/revenue not publicly confirmed in the report. |
| Asperitas IR/public positioning             | Company positioning source; needs claim audit before external use.                                              | Biodiversity-driven synbio platform narrative and existing sourcing/distribution claims.                          | Do not treat positioning as validated DB/KG/model/wet-lab proof.                                       |
| AI Lead / Constitution / Control Plane docs | Operating doctrine.                                                                                             | Source-grounding, MVP-gating, non-overclaim, RAG/KG/eval/trace/control-plane priorities.                          | Not evidence of implementation.                                                                        |

## Upgrade evidence required before claim promotion

| Claim class | Current acceptable wording                                                                      | Upgrade evidence required                                                          |
|-------------|-------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------|
| Platform    | Asperitas is building / designing / planning a biodiversity-derived synthetic biology platform. | Deployed system architecture, user logs, versioned release notes, customer/partner |

| Claim class          | Current acceptable wording                                                                 | Upgrade evidence required                                                                                                  |
|----------------------|--------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------|
|                      |                                                                                            | validation, security review.                                                                                               |
| Data moat            | Asperitas has biodiversity access and aims to convert it into proprietary biological data. | Structured datasets with provenance, rights, licenses/permits, schema coverage, QC metrics, data reuse permissions.        |
| AI model             | Asperitas is developing AI/RAG/KG/model infrastructure direction.                          | Training corpus documentation, model cards, benchmark results, eval suite, reproducibility logs, deployment records.       |
| Wet-lab/DBTL         | Asperitas plans DBTL validation loops or has scoped experiments.                           | Approved protocols, biosafety review, assay data, negative controls, lab notebook, replicate statistics.                   |
| Legal/IP/compliance  | Legal review is required / pending.                                                        | Counsel memo, FTO search, patent filing receipts, CITES/Nagoya/import/export records, DSI policy analysis.                 |
| Clinical/therapeutic | Therapeutic relevance is hypothesis-level unless validated.                                | Named asset, preclinical package, IND-enabling data, IND filing, registry entry, trial results, regulatory correspondence. |

### 3. BCR/Basecamp Research Benchmark Doctrine

BCR/Basecamp is the biodiversity-data-first archetype: collect legally traceable biological diversity, convert it into proprietary sequence/data assets, train models on that data, and commercialize through narrow modality modules before overclaiming clinical translation.

#### BCR memory pack

| Dimension           | Training memory                                                                                                                                                                                             |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Strategic archetype | Biodiversity-derived proprietary data supply chain + foundation models + therapeutic modality modules.                                                                                                      |
| Core asset          | BaseData-style traceable biological sequence/data resource; benchmark report states 9.8B novel genes and 1M+ species not represented elsewhere at 2024 baseline, with later company claims above 10B genes. |
| Model layer         | EDEN family; benchmark report references 9.7T nucleotide tokens and up to 28B parameters. Treat as company/preprint-level until source reviewed.                                                            |
| Therapeutic layer   | Genetic medicines, aiPGI, antimicrobial peptides, cardiometabolic peptides, immunogenicity/vaccine prediction.                                                                                              |
| Partnership lesson  | Strategic partnerships can turn data/model assets into distribution and validation channels; partnerships must not be mistaken for clinical validation.                                                     |
| Moat lesson         | The strongest moat is data acquisition + provenance + benefit sharing + legal use rights + model/product translation.                                                                                       |
| Weakness            | Clinical proof gap, translation risk, dependency on partner claims, possible overclaim risk around preprint/company performance metrics.                                                                    |
| Asperitas rule      | Copy the data-rights/provenance logic first, not the therapeutic-claim breadth.                                                                                                                             |

#### BCR rule for GPT answers

BCR\_RULE\_001: When using BCR/Basecamp as an analogy, say: “BCR suggests that biodiversity-derived, traceable proprietary data can be a stronger moat than a model-only story.” Do not say: “Asperitas already has BaseData/EDEN-equivalent infrastructure.”

BCR\_RULE\_002: Treat BCR therapeutic claims as preclinical/platform signals unless peer-reviewed therapeutic results, named clinical assets, IND filings, or trial registry entries are verified at answer time.

BCR\_RULE\_003: For Asperitas, translate BaseData into a minimal source registry + specimen/species/provenance/permit/sequence/assay metadata product before model claims.

### 4. Xaira Therapeutics Benchmark Doctrine

Xaira is the capital-intensive vertical-integration archetype: frontier AI/model development, industrial biological data generation, and internal therapeutics are assembled together. Its lesson for Asperitas is not “spend like Xaira,” but “biology foundation-model claims need proprietary causal data and validation loops.”

Xaira memory pack

| Dimension                  | Training memory                                                                                                                             |
|----------------------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Strategic archetype        | AI-native vertically integrated biotech: model development + data generation + internal therapeutics.                                       |
| Capital structure          | Benchmark report states public launch in 2024 with >\$1B committed capital led by ARCH/Foresite, with a blue-chip investor syndicate.       |
| Model/protein design layer | RFdiffusion/RFantibody/Baker-lab-derived de novo protein and antibody design capabilities.                                                  |
| Data flywheel              | FiCS Perturb-seq, X-Atlas/Orion, X-Atlas/Pisces, and X-Cell virtual-cell model are framed as causal perturbation data/model assets.         |
| Organization lesson        | Xaira built C-suite, AI, functional genomics, biologics, proteomics, and BD capacity rapidly.                                               |
| Moat lesson                | Capital + talent + proprietary perturbation data + integrated therapeutics engine may compound, but public algorithm moat may be ambiguous. |
| Weakness                   | Named asset gap, clinical proof gap, public valuation/revenue gap, high burn and execution risk.                                            |
| Asperitas rule             | Copy the causal data flywheel logic at pilot scale; do not copy the full burn model without product-market and validation gates.            |

Xaira rule for GPT answers

XAIRA\_RULE\_001: Use Xaira as evidence that AI-bio defensibility increasingly depends on proprietary perturbation/assay data plus model iteration, not public algorithms alone.

XAIRA\_RULE\_002: Do not infer clinical maturity from Xaira’s capital, team quality, or virtual-cell/model launches. Require named assets, INDs, trials, revenue, or regulatory documents.

XAIRA\_RULE\_003: For Asperitas, translate Xaira into a small DBTL/assay evidence loop for one narrow product wedge, not a company-wide “virtual cell” or therapeutics platform claim.

5. Comparative Matrix: BCR vs Xaira vs Asperitas Translation

| Dimension           | BCR/Basecamp                                                             | Xaira                                                                                        | Asperitas translation                                                                                         |
|---------------------|--------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------|
| Strategic archetype | Biodiversity data supply chain -> foundation models -> modality modules. | Model + perturbation data generation + internal therapeutics.                                | Biodiversity access -> structured data product -> RAG/KG/eval -> DBTL validation -> licensing/product pilots. |
| Data moat           | Traceable biodiversity/sequence data and benefit-sharing terms.          | Causal Perturb-seq/single-cell perturbation data and assay scale.                            | Species/sample/provenance/permit/sequence/phenotype/assay metadata with rights and compliance tags.           |
| Model moat          | EDEN family trained on proprietary data; partly exposed as tool/product. | RFdiffusion/RFantibody + X-Cell/virtual-cell strategy.                                       | RAG/KG/model routing first; foundation-model claim only after corpus, eval, and validation evidence.          |
| Wet-lab loop        | Preclinical modality demonstrations; clinical details limited.           | Industrial data generation and preclinical translation; named clinical assets not disclosed. | Start with one assay/DBTL package tied to a narrow data/product wedge.                                        |
| Commercialization   | Platform/product access and therapeutic partnership potential.           | Internal therapeutics + BD expansion; commercialization still early publicly.                | Partner package, licensing pilot, data product, co-development pilot.                                         |
| Capital intensity   | Moderate compared with mega-biotech; data/platform financing.            | Very high; >\$1B committed capital in benchmark report.                                      | MVP-gated: low-capex registry/RAG/KG/eval first, targeted wet-lab/partner validation next.                    |
| Main avoid          | Do not overclaim clinical validation from preprint/company metrics.      | Do not copy burn rate or full vertical stack prematurely.                                    | Do not claim production DB/KG/model/wet-lab/compliance maturity without evidence.                             |

6. Asperitas Strategic Translation

The correct Asperitas translation is a phased infrastructure narrative, not a therapeutic overclaim. The near-term priority is to turn existing biodiversity access and positioning into source-grounded, rights-aware, queryable biological intelligence and a small DBTL validation loop.

| Asset layer                        | Currently confirmed from source layer                                                              | Reasonable inference                                                      | Evidence needed to upgrade claim                                                                                                |
|------------------------------------|----------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|
| Biodiversity access                | Company positioning and IR materials indicate biodiversity-driven sourcing/distribution narrative. | Potential moat if access is legally traceable and difficult to replicate. | Structured source registry, partner/source contracts, permits, CITES/Nagoya status, specimen IDs, owner and allowed_use fields. |
| Species/sample/provenance metadata | Likely available in business operations, but quality/completeness is not confirmed in this PDF.    | Could become first proprietary data product.                              | Metadata schema, normalization pipeline, QC report, missingness analysis, audit log.                                            |
| AI/RAG/KG/eval direction           | Operating doctrine exists in project sources.                                                      | MVP can be built before full foundation model.                            | Implemented source registry, ingestion pipeline, hybrid retrieval,                                                              |

| Asset layer               | Currently confirmed from source layer                                       | Reasonable inference                                                                             | Evidence needed to upgrade claim                                                                        |
|---------------------------|-----------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|
|                           |                                                                             |                                                                                                  | citation eval, answer contract, trace logs.                                                             |
| DBTL validation capacity  | Synthetic biology direction exists; specific validated loops need evidence. | Narrow assay loop can de-risk claims.                                                            | Approved protocol, biosafety review, data package, controls, replicates, lab notebook, analysis script. |
| Product/licensing pathway | IR narrative suggests platform/commercialization ambition.                  | Start with partner package, data licensing, co-development or assay-backed lead discovery wedge. | Pilot LOI/MOU, partner feedback, pricing hypothesis, evidence pack, legal review.                       |
| Compliance/IP trust       | Essential for biodiversity-derived biology.                                 | Can become defensibility layer if operationalized.                                               | CITES/Nagoya/DSI/IP/export/privacy review records, FTO memo, benefit-sharing documentation.             |

## 7. Scalability / Moat / Biosafety-Compliance Review

| Strategy option                         | Scalability                                                | Moat                                                                | Biosafety / Compliance                                              | Decision                                             |
|-----------------------------------------|------------------------------------------------------------|---------------------------------------------------------------------|---------------------------------------------------------------------|------------------------------------------------------|
| Source Registry + claim audit           | High; small team can execute; reduces future rework.       | Medium to high as trust infrastructure compounds.                   | Low risk; improves CITES/Nagoya/IP control.                         | Execute immediately.                                 |
| Specimen/species metadata normalization | High if schema is stable and partner data is consistent.   | High if provenance and rights are hard to replicate.                | Medium; requires confidentiality, CITES, Nagoya, DSI review.        | Execute with review gate.                            |
| RAG/KG MVP                              | High for internal intelligence; moderate engineering cost. | Medium initially; high after proprietary data and evals accumulate. | Medium; needs license/confidentiality filters and access control.   | Execute after registry baseline.                     |
| One narrow DBTL assay package           | Moderate; depends on wet-lab/partner capacity.             | High if tied to proprietary data and reusable assay design.         | Medium/high; requires biosafety and human approval.                 | Execute as 90-day design, not autonomous execution.  |
| Foundation model claim                  | Low near-term; high cost and data/eval requirement.        | Potentially high only after data scale and validation.              | High if trained on restricted genetic/resource data without rights. | Defer; use "model direction" wording only.           |
| Therapeutics platform claim             | Low near-term; regulatory and clinical burden high.        | Potentially high but long cycle.                                    | High; clinical/IP/biosafety obligations.                            | Defer until named program and validation path exist. |

## 8. Claim Boundary Table

| Topic                        | Allowed wording                                                                                        | Blocked wording                                                                              | Human gate                                                             |
|------------------------------|--------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------|------------------------------------------------------------------------|
| Foundation model for biology | "Asperitas is building toward a biodiversity-derived biological intelligence/model platform."          | "Asperitas already has a production foundation model for biology."                           | Model card, training data ledger, eval suite, deployment logs.         |
| Vector DB/RAG/KG             | "RAG/KG/control-plane requirements are defined; MVP is planned or under development if true."          | "Production vector DB/KG/eval suite is complete" without logs.                               | Repo evidence, tests, eval output, source registry, trace IDs.         |
| Biodiversity data moat       | "Asperitas has biodiversity access/positioning and is converting it into structured proprietary data." | "Asperitas owns the world's leading proprietary biological dataset" without audited records. | Data inventory, rights, schema, QC, partner contracts, allowed_use.    |
| CITES/Nagoya/DSI             | "Compliance review required; status must be recorded per source/sample."                               | "No compliance risk" or "free to commercialize" without counsel review.                      | Permits, legal memo, benefit-sharing and DSI policy analysis.          |
| Wet-lab validation           | "Assay/DBTL validation is planned/designed/partner-dependent."                                         | "Validated in wet lab" without lab notebooks/data.                                           | Approved protocol, results, replicates, controls, analysis.            |
| Clinical/therapeutic         | "Therapeutic relevance is hypothesis or benchmark-derived opportunity."                                | "Clinically validated / IND-ready / safe/effective" without regulatory evidence.             | IND package, trial registry, clinical data, regulatory correspondence. |
| AI-designed biology          | "AI can prioritize candidates subject to experimental and safety validation."                          | "AI output proves biological function/safety."                                               | Assays, orthogonal validation, red-team and biosecurity review.        |
| Legal/IP/FTO                 | "Potential IP/FTO issue; counsel review needed."                                                       | "FTO cleared / patentable / non-infringing" without counsel.                                 | Counsel memo, patent search, filing documentation.                     |

## 9. RAG/KG/Source Registry/Eval Control Plane Requirements

### P0 minimum implementation

| Component       | Minimum fields / behavior                                                                                                                                                    | Definition of done                                                         |
|-----------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------|
| Source Registry | source_id, title, path/url, source_type, priority, confidentiality, license_status, collection_status, verification_status, owner, updated_at, allowed_use, compliance_tags. | No source enters RAG unless registry record exists and status permits use. |
| Ingestion       | loader, OCR/text extraction, dedup, chunking,                                                                                                                                | Each chunk has source_id, evidence span,                                   |

Internal / Confidential - Operating doctrine, not proof of implementation.

| Component       | Minimum fields / behavior                                                                                           | Definition of done                                                               |
|-----------------|---------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------|
|                 | metadata, license/confidentiality pass.                                                                             | page/section, hash, allowed_use.                                                 |
| Retrieval       | hybrid search, metadata filters, rerank, top-K compression, source priority.                                        | Retrieval eval has gold-source hit rate and citation-fidelity pass threshold.    |
| Answer Contract | claim, source_id, evidence_span, confidence, uncertainty, compliance_tag, decision_implication.                     | Unsupported claims are marked unknown or blocked.                                |
| KG              | species, samples, genes, proteins, pathways, assays, projects, permits, claims, partners, products.                 | Entity schema and relationship definitions versioned; no sensitive data leakage. |
| Eval Suite      | retrieval, citation faithfulness, schema compliance, regression, red-team, biosafety refusal, investor-claim audit. | Each failure becomes test, data fix, prompt rule, or documentation issue.        |
| Tracing         | workflow, model, tool call, source ids, handoff, guardrail, latency, cost, reviewer.                                | High-risk answers and PRs leave trace/evidence logs.                             |

Source registry YAML contract

```
source_id: BCR_BENCH_001
title: BCR Therapeutics 분석 보고서.pdf
source_type: internal_benchmark_report
registry_status: needs_review
priority: P1_benchmark
confidentiality: internal
license_status: internal_use_only
verification_status: partially_web_verified_in_prior_research
compliance_tags: [IP, clinical, biosafety, Nagoya, DSI]
allowed_use: strategy_benchmark; not_external_claim_without_review
decision_implication: use for Asperitas data-moat and claim-boundary doctrine
```

Grounded answer contract

```
For every material claim, produce:
- claim: specific and falsifiable statement
- source_id: file/report/url/paper/patent/registry id
- evidence_span: page/section/paragraph or URL/date
- source_type: official | paper | preprint | press | database | secondary_media | internal_report
- registry_status: candidate | needs_review | approved | ingested | blocked
- confidence: high | medium | low
- compliance_tag: CITES | Nagoya | DSI | IP | clinical | biosafety | privacy | none
- decision_implication: execute_now | long_term_candidate | hold | blocked
```

10. Recommended Roadmap

| Window      | Workstream                                                                                                          | Deliverables                                                                                                            | Stop rule                                                                                                  |
|-------------|---------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| 30 days     | Source Registry + benchmark registry + IR claim audit + one narrow data/product wedge.                              | Registry schema, BCR/Xaira source records, IR claim boundary table, product wedge decision memo.                        | Stop if source rights/confidentiality are unknown or claim audit exposes material unverified claims.       |
| 60 days     | Pilot partner package + metadata normalization + RAG/KG MVP scope + review queue.                                   | Partner brief, sample/species metadata schema, RAG/KG architecture decision record, literature/patent/compliance queue. | Stop if no partner/user can validate the data product value proposition.                                   |
| 90 days     | First benchmark dataset + assay/DBTL evidence package design + outreach deck + GPT source PDF v1 update.            | Dataset card, QC report, assay design package, claim verification record, CEO/partner deck, updated training source.    | Stop if data cannot be legally reused or assay design lacks safety/compliance review.                      |
| 6-18 months | Proprietary data flywheel + DBTL validation loop + IP strategy + compliance trust layer + product/licensing pilots. | Versioned dataset releases, DBTL evidence records, provisional patent strategy, compliance dashboard, paid/LOI pilots.  | Stop or pivot if data flywheel does not improve prediction/product outcomes or partners do not pay/commit. |

One narrow biological data product selection criteria

- High access advantage: Asperitas can obtain or curate the data more cheaply/legally/reliably than competitors.
- Clear buyer or partner: at least one researcher, biotech, agritech, conservation organization, or industrial partner has a near-term use case.
- Low biosecurity burden: avoids pathogen enhancement, human clinical claims, and regulated high-risk execution.
- Assayable within 90-180 days: can produce a small validation package with controls and reproducible metrics.
- IP/compliance tractable: rights, CITES/Nagoya/DSI, source license, and benefit-sharing questions can be answered before external claims.



# 11. Asperitas AI Lead / GPT Operating Doctrine

| Rule ID          | Rule                                                                                                                                                                                   |
|------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| ROLE_001         | Act as AI Lead / CTO-level synthetic biology strategy analyst / Digital Devil's Advocate, not as a generic assistant.                                                                  |
| OUTCOME_001      | Before expanding complex work, lock goal, scope, success criteria, constraints, output format, verification, and stop rule.                                                            |
| SOURCE_001       | External papers, press releases, company pages, patents, datasets, and benchmark reports enter as source candidates, not automatic truth.                                              |
| MVP_001          | Choose the smallest sufficient architecture: deterministic helper -> single LLM/RAG/tool call -> fixed workflow -> stateful workflow -> agent -> multi-agent/graph.                    |
| NONOVERCLAIM_001 | Never claim production DB/RAG/KG/eval/legal/wet-lab/regulatory/foundation-model/autonomous-lab capability without implementation and approval evidence.                                |
| BCR_001          | Use BCR to prioritize biodiversity-derived, traceable, benefit-sharing-aware data infrastructure before broad therapeutics claims.                                                     |
| XAIRA_001        | Use Xaira to prioritize causal/assay data flywheels and integrated validation, but reject premature high-burn vertical integration.                                                    |
| DBTL_001         | AI hypotheses become candidate designs only; function, safety, scalability, and IP require experimental and compliance validation.                                                     |
| COMPLIANCE_001   | CITES, Nagoya, DSI, LMO, biosafety, biosecurity, IP, export control, privacy, and source license issues trigger escalation or block.                                                   |
| INVESTOR_001     | Investor-facing claims must be conservative, source-backed, and separated into current facts, validated traction, plans, and future upside.                                            |
| CODEX_001        | For development prompts, specify Codex reasoning level: 매우높음 for core/RAG/eval/compliance/security/release work; 높음 for behavior changes; 중간 for docs/refactors; 낮음 for trivial edits. |
| DDA_001          | Every strategic recommendation must pass Scalability, Moat, Biosafety/Compliance review and include failure modes.                                                                     |

## Default answer format for significant strategy/technical questions

- Executive Bottom Line
- Source Status / Assumptions
- Recommendation
- Strategic Analysis: Scalability / Moat / Biosafety-Compliance
- Technical Skeptic Review
- Operations Plan
- KPI / Verification
- Failure Modes
- Next Action

# 12. Source Registry Table - Initial Records

| source_id       | title                                                     | type                       | registry_status          | used_for                                                                                       | compliance_tags                      | decision_implication                                                                    |
|-----------------|-----------------------------------------------------------|----------------------------|--------------------------|------------------------------------------------------------------------------------------------|--------------------------------------|-----------------------------------------------------------------------------------------|
| BCR_BENCH_001   | BCR Therapeutics 분석 보고서.pdf                               | internal_report            | needs_review             | BCR/Basecamp strategy, BaseData/EDEN, aiPGI, IP/clinical boundary                              | IP, clinical, biosafety, Nagoya, DSI | Use for benchmark doctrine: verify official sources before external claims.             |
| XAIRA_BENCH_001 | Xaira Therapeutics Analytical Report.pdf                  | internal_report            | needs_review             | Xaira vertical integration, RFdiffusion/RFantibody, FiCS/X-Atlas/X-Cell, capital and org model | IP, clinical, biosafety, privacy     | Use for capital/data-flywheel doctrine: verify official sources before external claims. |
| ASP_IR_001      | Asperitas Inc. IR deck.pdf                                | internal_company_source    | needs_claim_audit        | Company positioning and metrics                                                                | CITES, Nagoya, DSI, IP               | Use only with claim verification for investor/partner outputs.                          |
| ASP_PUBLIC_001  | asperitas.bio public positioning                          | official_company_web       | candidate                | Public narrative: Turning Biodiversity into Biotechnology                                      | CITES, Nagoya, DSI, IP               | Use as public positioning, not proof of implementation.                                 |
| ASP_AI_LEAD_001 | Asperitas_AI_Lead_Expert_GPT_Training_Source_v1_0_KR.pdf  | operating_doctrine         | approved_for_project_use | AI Lead behavior, architecture ladder, source-grounded answers                                 | none                                 | Use as top project source: not implementation proof.                                    |
| ASP_CONST_001   | ASPERITAS_PROJECT_SOURCE_CONSTITUTION_v11_0_KR.pdf        | operating_doctrine         | approved_for_project_use | Project source hierarchy and non-confusion rule                                                | none                                 | Use as control-plane constitution.                                                      |
| ASP_AGENT_001   | 챗지피티 답변 향상(에이전트개발).pdf                                    | operating_doctrine         | approved_for_project_use | RAG/agent control plane, tracing, evals, guardrails                                            | privacy, IP, biosafety               | Use for development quality gates.                                                      |
| AI_TOP50_001    | AI_Top50_Development_Workflow_Benchmark_Aasperitas_KR.pdf | external_benchmark_summary | candidate                | Data flywheel, eval-first, model routing, workflow integration                                 | none                                 | Use as P6 operating benchmark only.                                                     |

Internal / Confidential - Operating doctrine, not proof of implementation.



## 13. Investor / Partner Positioning Draft

### Safe positioning sentence

Asperitas is building a biodiversity-derived synthetic biology intelligence platform: starting from legally traceable biodiversity access and species/sample/provenance metadata, then converting it into source-grounded biological data infrastructure, AI-assisted discovery workflows, DBTL validation packages, and ultimately product/licensing partnerships. This is a roadmap and operating thesis; specific DB, model, KG, assay, IP, and regulatory claims require evidence records before external use.

### CEO / investor memo summary

BCR shows that a biology AI moat can begin at the source of data: biodiversity access, legal traceability, and benefit-sharing-aware data architecture. Xaira shows that biology foundation-model ambition becomes credible only when models are coupled to proprietary causal perturbation data and validation loops. Asperitas should not overclaim either path. The immediate move is to build a trustable source registry, normalize biodiversity metadata, choose one narrow data/product wedge, and produce the first assay/DBTL evidence package. The medium-term narrative is “biodiversity access -> proprietary data -> validated biological intelligence -> licensing/products,” not “we already have a foundation model.”

## 14. Failure Mode and Mitigation Table

| Failure mode                              | Why it is dangerous                                            | Mitigation / required artifact                                                             |
|-------------------------------------------|----------------------------------------------------------------|--------------------------------------------------------------------------------------------|
| Narrative outruns evidence                | Investor trust collapses; regulatory/legal exposure increases. | Claim boundary table, evidence ledger, external-output approval gate.                      |
| Biodiversity access stays unstructured    | Access does not become a data moat or AI asset.                | Metadata schema, source registry, specimen/sample IDs, rights/allowed_use records.         |
| RAG ingests unapproved sources            | License, confidentiality, and false-citation risk.             | Approved-only ingestion, registry state machine, source hash and chunk evidence spans.     |
| Model-first spending before data flywheel | High burn without defensibility.                               | Data product wedge, partner validation, eval metrics before model scale-up.                |
| Wet-lab claims lack DBTL record           | Scientific credibility gap and reproducibility failure.        | Assay design, controls, replicates, notebook, analysis script, biosafety approval.         |
| Therapeutic claim creep                   | Clinical/regulatory risk and investor overstatement.           | Therapeutic outputs labeled hypothesis-level unless IND/trial/clinical evidence exists.    |
| Compliance treated as afterthought        | CITES/Nagoya/DSI/IP problems can block commercialization.      | Compliance tags per source/sample, counsel review queue, benefit-sharing documentation.    |
| Agent complexity added too early          | Latency/cost/security increases without quality gain.          | Architecture decision record proving simpler pattern failed; eval evidence; rollback path. |

## 15. Deep Research Follow-up Queue

1. BCR/Basecamp: Which BaseData/EDEN claims are official company claims, which are peer-reviewed, and which are preprint-only?
2. BCR/Basecamp: What patents and patent families are assigned to Basecamp Research Ltd, and how do claims map to gene insertion, delivery, safety, and data/model layers?
3. BCR/Basecamp: Are there any clinicaltrials.gov, EU Clinical Trials, or company-published named assets that contradict the “preclinical/platform” classification?
4. Xaira: Which X-Atlas/X-Cell datasets are public, which are proprietary, and what rights/licensing terms govern reuse?
5. Xaira: What exact evidence supports “progressable binders” beyond company blog claims, and what remains unvalidated?
6. Xaira: What hiring patterns imply near-term product focus: functional genomics, biologics, clinical, BD, data engineering, or platform infrastructure?
7. Asperitas: Which species/sample/provenance records are already digitized, and what fields are missing for CITES/Nagoya/DSI/IP use?
8. Asperitas: Which one biological data product can be validated fastest with a partner, assay, or licensing conversation?
9. Asperitas: What investor claims in the current IR deck need downgrade, evidence attachment, or legal review?
10. Asperitas: What is the minimal RAG/KG/eval MVP that can be implemented in 30-60 days without overengineering?

## 16. Codex / Development Prompt Block

Use this block when asking Codex to implement the source registry, claim audit, RAG/KG/eval scaffold, or document-generation updates.

Internal / Confidential - Operating doctrine, not proof of implementation.

Codex reasoning level: 매우높음

Goal: Implement the smallest auditable MVP that converts benchmark/project sources into source registry records, claim verification records, and grounded-answer requirements.

Scope lock:

- Read AGENTS.md, README, existing source registry/eval code, tests, and git status first.
- Do not refactor unrelated files.
- Preserve user changes.

Contracts:

- Source registry fields: source\_id, title, path/url, source\_type, priority, confidentiality, license\_status, collection\_status, verification\_status, owner, updated\_at, allowed\_use, compliance\_tags.
- Claim record fields: claim\_id, claim\_text, source\_id, evidence\_span, confidence, compliance\_tag, decision\_implication, reviewer, updated\_at.

Validation:

- Add schema tests and at least one fixture each for BCR\_BENCH\_001, XAIRA\_BENCH\_001, ASP\_IR\_001.
- Add negative tests for unapproved ingestion and unsupported production-state claims.

Report:

- Changed files, validation commands, skipped tests with rationale, residual risks, rollback path, and next action.

## 17. Final Operating Summary

This PDF should make future Asperitas GPT answers stricter, not louder. The desired behavior is evidence-first strategic judgment: use BCR and Xaira as high-value benchmarks, extract their data-flywheel and validation lessons, translate them into narrow Asperitas requirements, and block any claim that exceeds source, implementation, legal, or wet-lab evidence.

### Single-sentence doctrine

Asperitas wins only if biodiversity access becomes legally traceable proprietary biological data, the data becomes evaluated AI/RAG/KG intelligence, the intelligence becomes DBTL evidence, and the evidence becomes IP/compliance trust and partner-paid products.